

# Chamudi Siriwardhane

## Machine Learning Enthusiast

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### About Me

Fourth-year Information Systems undergraduate with a strong foundation in data science, machine learning, and AI technologies. Passionate about leveraging statistical analysis and predictive modeling to derive actionable insights from complex data. Seeking opportunities to apply my technical expertise in Python, SQL, and machine learning frameworks to solve real-world business challenges and drive data-informed decision making.

### Experience

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| <b>Software Engineer Intern</b> , Entgra – Sri Lanka                                                                                                                                                                                                                                                                                                                                                                                        | 2024 - 2025 |
| <ul style="list-style-type: none"><li>Automated workflows using shell scripts reducing manual effort</li><li>Developed RESTful APIs</li><li>Implemented bug fixes and optimizations</li><li>Gained hands-on experience with full development lifecycle and Linux environments</li></ul> <b>Tools &amp; Technologies:</b> Java, React, SQL, Swagger, Postman, Linux, Git                                                                     |             |
| <b>Software Engineer (Volunteer)</b> , Dev Team AIESEC Sri Lanka                                                                                                                                                                                                                                                                                                                                                                            | 2024 - 2026 |
| <ul style="list-style-type: none"><li>Implemented UTM tracking analytics system to enable data-driven marketing campaign decisions</li><li>Designed and developed user-friendly interfaces with focus on accessibility and user experience</li><li>Automated data migration and entry workflows using Selenium and Python.</li></ul> <b>Tools &amp; Technologies:</b> JavaScript, Node.js, MongoDB, Postman, Git, Python, ClickUp, Selenium |             |

### Education

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|------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>University of Colombo School of Computing</b> , B.Sc (Hons) in Information Systems                                        | 2022 – 2026 |
| <ul style="list-style-type: none"><li>CGPA: 3.42/4.0</li></ul>                                                               |             |
| <b>Anula Vidyalaya Nugegoda</b>                                                                                              | 2010 – 2020 |
| <ul style="list-style-type: none"><li>G.C.E. Advanced Level (2020): A - ICT, B - Combined Mathematics, S - Physics</li></ul> |             |

### Projects

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| <b>Cognitive Distortion Classification for CBT Delivery with XAI Integration (Final Year Research)</b>                                                                                                                                                                                                                                                                               | 2025 – Present |
| <ul style="list-style-type: none"><li>Developing a Python-based model for classifying cognitive distortions into five categories</li><li>Using XAI techniques (SHAP and LIME) to provide interpretable explanations for clinicians as a CBT delivery aid</li></ul> <b>Tools &amp; Technologies:</b> Python, scikit-learn, SHAP, LIME, pandas, NumPy, Jupyter Notebook, Google Collab |                |
| <b>Wine Dataset Clustering Using K-Means and Hierarchical Methods</b>                                                                                                                                                                                                                                                                                                                | 2025           |
| <ul style="list-style-type: none"><li>Conducted unsupervised clustering on wine datasets using <b>K-Means</b> (with Elbow method for optimal cluster selection) and <b>Hierarchical Clustering</b> (agglomerative approach) to identify natural groupings.</li></ul> <b>Tools &amp; Technologies:</b> R programming, RStudio                                                         |                |
| <b>Appliance Energy Consumption Prediction Using Multivariate Time-Series Data</b>                                                                                                                                                                                                                                                                                                   | 2025           |
| <ul style="list-style-type: none"><li>Preprocessed and analyzed a multivariate appliance energy dataset recorded at 10-minute intervals, including environmental and time-based features.</li><li>Engineered meaningful features and applied deep learning techniques to predict future energy consumption accurately.</li></ul>                                                     |                |

- Evaluated model performance and refined architecture for improved prediction of energy usage patterns.

**Tools & Technologies:** Python, Pandas, NumPy, TensorFlow, Keras, Jupyter Notebook

### Market Basket Analysis Using Association Rule Mining

2025

- Conducted Market Basket Analysis on retail transaction data to uncover customer spending patterns using association rule mining techniques.
- Applied support, confidence, and lift metrics to identify frequent itemsets and generate actionable insights for targeted promotions.
- Implemented the analysis using Weka software, exploring data preprocessing, rule generation, and visualization of customer purchasing behavior.

**Tools & Technologies:** Weka, Microsoft Excel, Data Visualization

## Programming Languages and Tools

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**Programming Languages:** Python, R, SQL, Java

**Data Science & ML Tools:** Pandas, NumPy

**Data Visualization:** Power BI, Tableau, Matplotlib, Seaborn

**Large Language Models:** OpenAI API

**Database Management:** MongoDB, MySQL

**Development:** Git, Docker, Linux, REST APIs

**Web Technologies:** JavaScript, ReactJS, HTML/CSS

## Skills

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- Programming
- Data Analysis
- Data Visualization
- Development
- Database
- Machine Learning
- NLP

## Honors & Awards

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**Director's List - 2023 (AY 20)** (Issued by University of Colombo School of Computing)

Oct 2024

## Volunteering

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- IEEE WIE Student Branch Affinity Group of UCSC
- AIESEC Sri Lanka
- AIESEC in University of Colombo
- ACM Student Chapter of UCSC